

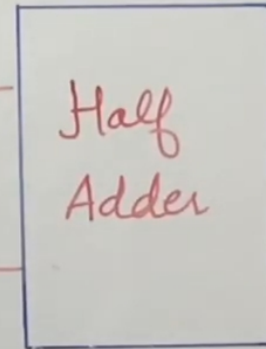
Designing of a Half Adder

* A logic ckt. for the addition of two one bit numbers is called Half Adder

2^n
 $n \rightarrow \text{no. of I/Os}$
 $2^2 = 4$

I/Ps		S
A	B	
0	0	0
0	1	1
1	0	1
1	1	0

A (one bit no.)



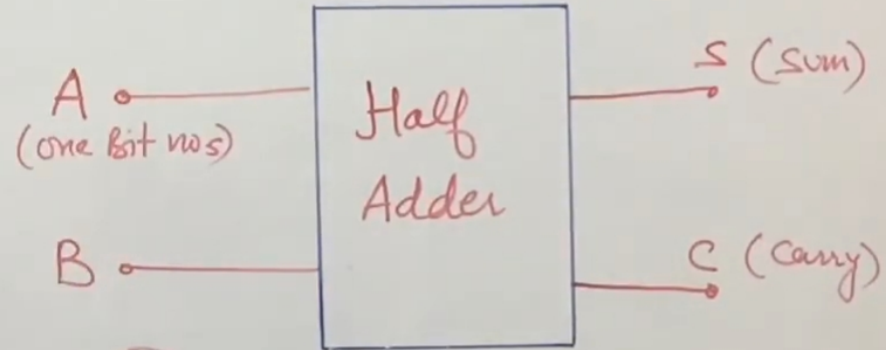
0	1
1	0

$$S = A\bar{B} + \bar{A}B = A \oplus B$$
$$C = A \cdot B$$

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I/Ps		O/Ps	
A	B	$\check{S}$	$\check{C}$
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



		S	
A	B	0	1
0	0	0	1
1	0	1	0

$$S = A\bar{B} + \bar{A}B = \underline{A \oplus B}$$

$$C = \underline{A \cdot B}$$

